Chapter 3 Sampling Proportions and Percentages

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Qualitative characteristics

Wish to estimate the total number, the proportion, or percentage of units in the population that possess some characteristic or attribute or fall into some defined class.

- number of unemployed persons
- the proportion of a “yes” answer to a question
- the percentage of the population that is native-born
- the percentage of the population aged 60 and over

Notation

Suppose: every unit in the population falls into one of the two classes C and C'.

Number of units in C		Proportion of units in C	
Population	Sample	Population	Sample
A	a	$P = A/N$	$p = a/n$

- The sample estimate of P is p, $\hat{P} = p$.
- The sample estimate of A is $\hat{A} = Np = Na/n$.

Variances of the sampling estimates

For any unit in the sample or population

$$y_i = \begin{cases} 1, & \text{if the unit } i \text{ is in C} \\ 0, & \text{if it is in C}' \end{cases}$$

- $Y = \sum_{i=1}^{N} y_i = A$
- $\bar{Y} = \frac{1}{N}\sum_{i=1}^{N} y_i = \frac{A}{N} = P$
- $\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{a}{n} = p$

* The characteristics A and P are special forms of the total and mean of a population.

Theorem 3.1 The sample proportion $p=a/n$ is an unbiased estimate of the population proportion $P = A/N$.

Note that $\sum_{i=1}^{N} y_i^2 = A = NP$ $\sum_{i=1}^{n} y_i^2 = a = np$

Hence

$$S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - \bar{Y})^2 = \frac{\sum_{i=1}^{N} y_i^2 - N\bar{Y}^2}{N-1} = \frac{1}{N-1}(NP - NP^2) = \frac{N}{N-1}PQ$$

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 = \frac{n}{n-1}pq$$

Theorem 3.2. The variance of p is

$$V(p) = E(p-P)^2 = \frac{S^2}{n}\frac{(N-n)}{N} = \frac{PQ}{n}\left(\frac{N-n}{N-1}\right)$$

Corollary 1. If p and P are the sample and population *percentages*, respectively, falling into class C, the variance of p is the same as above.

Corollary 2. The variance of $\hat{A} = Np$, the estimated total number of unit in class C, is

$$V(\hat{A}) = \frac{N^2 PQ}{n}\left(\frac{N-n}{N-1}\right)$$

Theorem 3.3. An unbiased estimate of the variance of p, derived from the sample, is

$$v(p) = s_p^2 = \frac{N-n}{(n-1)N} pq$$

Corollary. An unbiased estimate of the variance of $\hat{A} = Np$, the estimated total number of unit in class C, is

$$v(\hat{A}) = s_{Np}^2 = \frac{N(N-n)}{n-1} pq$$



Example

From a list of 3042 names and addresses, a simple random sample of 200 names showed on investigation 38 wrong addresses. Estimate the total number of addresses needing correction in the list and find the standard error of this estimate.

We have $N = 3042,\ n = 200,\ a = 38,\ p = 0.19$

The estimated total number of wrong addresses is

$$\hat{A} = Np = 3042 \times 0.19 \approx 578$$

$$v(\hat{A}) = \frac{N(N-n)}{n-1} pq = \frac{3042 \times (3042-200)}{200-1} \times 0.19 \times 0.81 \approx 6686$$

$$se(\hat{A}) = \sqrt{6686} = 81.8$$

Confidence limits

The Normal approximation

One form of the normal approximation to the confidence limits for P is

$$p \pm \left[t\sqrt{1-f}\,\sqrt{pq/(n-1)} + \frac{1}{2n} \right]$$

- A correction for continuity: a slight improvement
- Without the correction, the normal approximation usually gives too narrow a confidence interval.
- The error in the normal approximation depends on all the quantities $n,\ p,\ N,\ \alpha_U,\ \alpha_L$.

- The quantity to which the error is most sensitive is the number observed in the smaller class.

SMALLEST VALUES OF np FOR USE OF THE NORMAL APPROXIMATION

p	np = Number Observed in the *Smaller* Class	n = Sample Size
0.5	15	30
0.4	20	50
0.3	24	80
0.2	40	200
0.1	60	600
0.05	70	1400
~0*	80	∞

* This means that p is extremely small, so that np follows the Poisson distribution.

Rules: with 95% confidence limits the true frequency with which the limits fail to enclose P is not greater than 5.5%.



Example

In a simple random sample of size 100, from a population of size 500, there are 37 units in class C. Find the 95% confidence limits for the proportion and for the total number in class C in the population. In this example

$$n = 100, \qquad N = 500, \qquad p = 0.37$$

The example lies in the range in which the normal approximation is recommended. The estimated standard error of p is

$$\sqrt{(1-f)pq/(n-1)} = \sqrt{(1\text{-}100/500) \times 0.37 \times 0.63/99} = 0.434$$

The correction for continuity, $1/2n$, equals 0.005.

Hence the 95% limits for P are estimated as

$$0.37 \pm (1.96 \times 0.0434 + 0.005) = 0.37 \pm 0.090$$

$$\hat{P}_L = 0.280, \qquad \hat{P}_U = 0.460$$

To find limits for the total number in class C in the population, we multiply by N, obtaining 140 and 230.

The hypergeometric distribution

- The numbers of units in the two classes C and C’ in the population are A and A'.
- The probability that the sample contains a units in C and a' units in C’ is

$$P(a,a' \mid A,A') = \binom{A}{a}\binom{A'}{a'} \Big/ \binom{N}{n}$$

- Hypergeometric distribution: the frequency distribution of a or np
- The distribution of p is immediately derivable.



Example

A family of eight contains three males and five females. Find the frequency distribution of the number of males in a simple random sample of size 4. In this case

$$A = 3;\quad A' = 5,\qquad N = 8;\quad n = 4$$

The distribution of the number of males, a, is as follows

a	0	1	2	3	4
Prob	1/14	6/14	6/14	1/14	0



p	0	1/4	1/2	3/4
Prob	1/14	6/14	6/14	1/14

$$E(p) = \frac{1}{4}\times\frac{6}{14} + \frac{1}{2}\times\frac{6}{14} + \frac{3}{4}\times\frac{1}{14} = \frac{3}{8} \qquad E(p) = P = \frac{A}{N} = \frac{3}{8}$$

$$V(p) = \left(0\text{-}\frac{3}{8}\right)^2 \times \frac{1}{14} + \cdots = \frac{15}{448} \qquad V(p) = \frac{PQ}{n}\frac{N-n}{N-1} = \frac{15}{448}$$

Classification into more than two classes

Suppose: the number falling in the ith class is A_i in the population and a_i in the sample.

$$N = \sum A_i, \quad n = \sum a_i, \quad P_i = \frac{A_i}{N}, \quad p_i = \frac{a_i}{n}$$

- Case 1: we calculate

$$p = \frac{\text{number in any one class in sample}}{n} = \frac{a_1}{n}$$

or

$$p = \frac{\text{total number in a group of classes}}{n} = \frac{a_1 + a_2}{n}$$

Basically, p is obtained from a subdivision of the n units into only two classes.

- Case 2: Sometimes certain classes are omitted, p being computed from a breakdown of the remaining classes into two parts.

Suppose $p = \dfrac{a_1}{a_1 + a_2}, \quad n' = a_1 + a_2, \quad n = a_1 + a_2 + a_3, \quad N' = A_1 + A_2$

where a_3 is the number in the sample falling in classes in which we are not at the moment interested.

The normal approximations to conditional confidence limits for $P = A_1 / (A_1 + A_2)$ are

$$p \pm \left[t \sqrt{\left(1 - \frac{n'}{N'}\right) \frac{pq}{(n' - 1)}} + \frac{1}{2n'} \right]$$

Proportions and totals over subpopulations

The sample data may be presented as follows.

	Domain 1 N_1		Domain 2 N_2		… …	Domain k N_k		Total N
Class	C	C'	C	C'	…	C	C'	
Number of units	A_1	A_1'	A_2	A_2'	…	A_k	A_k'	N
	a_1	a_1'	a_2	a_2'		a_k	a_k'	n

- The proportion falling in class C in domain 1 is estimated by

$$p_1 = a_1 / (a_1 + a_1')$$

- With $n_1 = a_1 + a_1'$, the standard error is

$$s(p_1) = \sqrt{\left(1 - \frac{n_1}{N_1}\right)\left(\frac{p_1 q_1}{n_1 - 1}\right)}$$

Estimate the total number A_1 of units in class C in domain 1

- If the total number of units in domain 1 in the population N_1 is **known**,

$$\hat{A}_1 = N_1 p_1 = \frac{N_1 a_1}{a_1 + a_1'}$$

The estimated standard error is, ($n_1 = a_1 + a_1'$)

$$s(\hat{A}_1) = N_1 \sqrt{1-(n_1 / N_1)} \sqrt{p_1 q_1 / (n_1 - 1)}$$

- If N_1 is **unknown**,

$$\hat{A}_1' = \frac{N a_1}{n} \triangleq Np$$

The estimated standard error is

$$s(\hat{A}_1) = N \sqrt{1-(n / N)} \sqrt{pq / (n-1)}$$